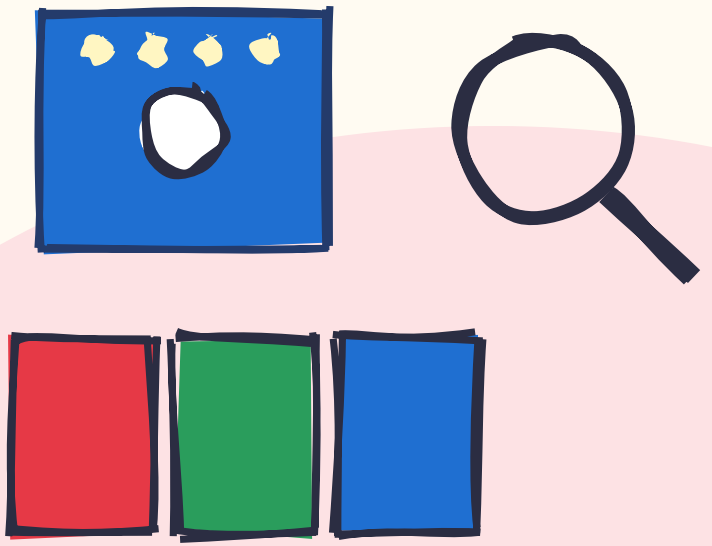
Little Explorers · Class 23

Colour Detective Sensor

Teach a computer to see colours!

Light & colour | Colour sensor | Sorting



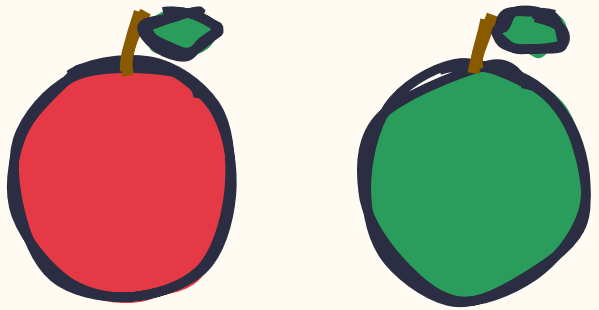
Did you know?
Recycling factories use colour sensors to sort green, brown and clear glass bottles super fast.

Let's wonder...

Can a computer see colours?

A colour mystery

You can tell a red apple from a green leaf in a second. But a computer has no eyes! How could it ever know which colour something is?



It can't!

With a camera?

With a colour sensor!

Today: you become **colour detectives** and use a sensor that tells red, green and blue apart.

LEARN

Learning objectives

Today I can...

1	**See colour:** say that we see colour because **light bounces** off things	**Proof:** I explained why a banana looks yellow
2	**Sensor detective:** use a **colour sensor** to name red, green and blue cards	**Proof:** the sensor named 3 cards correctly
3	**Sort it:** **sort** objects by colour, like a recycling robot	**Proof:** our team sorted 6 objects

Objective 1 · See colour · video

Video time!



The Science of Light and Color for Kids

Free School · about 5 minutes · youtu.be/9Vsl0lom3S0

Watch for...

1. Why can we see **colours**?
2. What happens to light when it hits a red apple?
3. Why is a rainbow made of many colours?

Scan to watch

After the video: Look around the room: find something red, something green and something blue!

Objective 1 · See colour

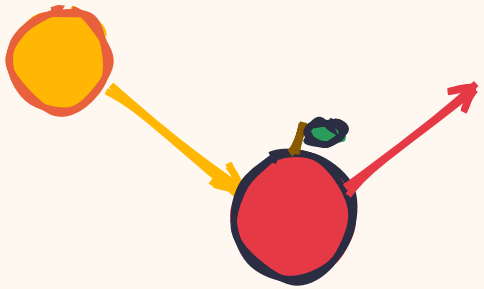
Colour comes from light

The idea

Light from the sun has **all** the colours mixed together. When it hits a red apple, the apple keeps most colours and **bounces back red** into your eyes.

No light = no colour. In the dark, everything looks black!

Example: a ball bouncing



Throw a ball at a wall: it bounces back to you. Light bounces off things too, and the bounced light carries the colour to your eyes.

Objective 2 · Sensor detective

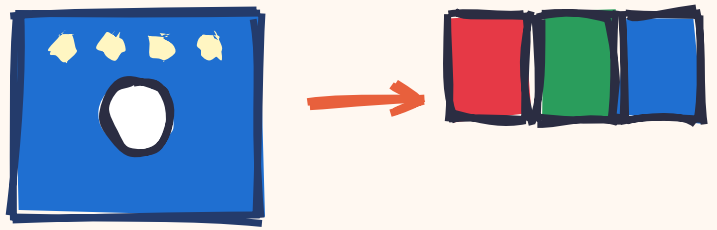
How the colour sensor works

The idea

The sensor has its own tiny **lights** to shine on the card. Then it **measures** how much red, how much green and how much blue bounces back.

Lots of red bounces back? It's a **red** card!

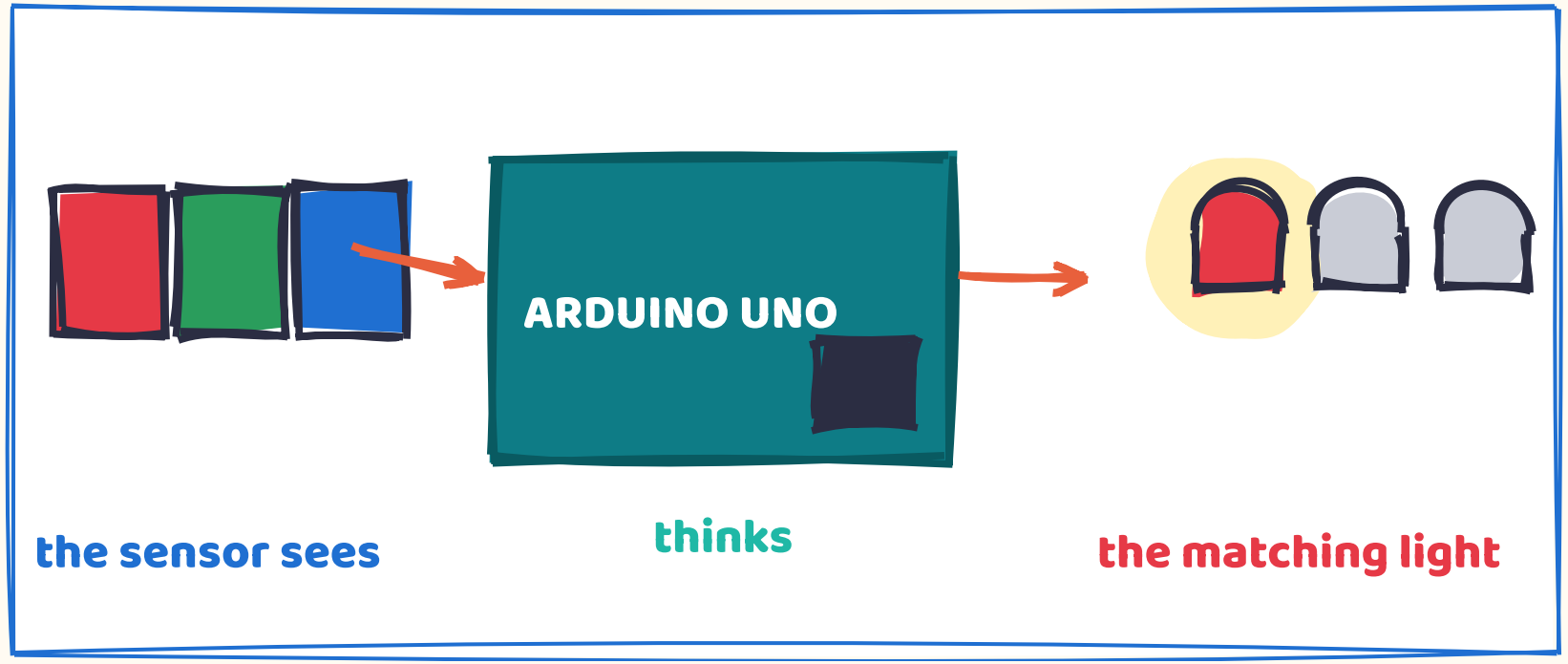
Example: tasting food



You taste a drink and say 'mostly orange, a bit of mango'. The sensor 'tastes' light and says 'mostly red, a bit of blue'.

Our machine

See, think, show



How it works

The **sensor** measures the card.
The Arduino **thinks**: which colour is biggest?
The **matching light** turns on and the screen says the colour!

Teacher: parts are pre-wired on the breadboard. Children plug in the USB cable, snap the picture blocks in mBlock and test.

Objective 3 · Sort it

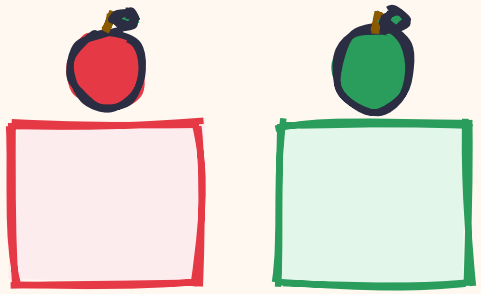
Sorting robots

The idea

Once a computer knows a colour, it can **sort**: red things in this box, green things in that box.

Factories sort sweets, fruit, and **recycling** with colour sensors. Faster than people!

Example: sorting laundry



Before washing, we sort clothes: whites in one pile, colours in another. A sorting robot does the same job, but never gets bored!

Real world

Colour sensors at work



Recycling
Sort glass and plastic by colour



Fruit factories
Find ripe and unripe fruit



Phones
Change screen colours for the room's light



Paint shops
Match any paint colour exactly

? What would you like a colour-sorting robot to sort at home?

Fun fact
Some sweet factories check every single sweet's colour with a sensor!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Lights out!
2. What the computer is thinking
3. Detective tests
4. Be the sorting robot
5. Detective challenges

Class game

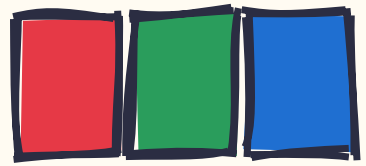
Lights out!

How to play

1. Close the curtains and turn off the lights.
2. Hold up coloured cards. Can you name them?
3. Now shine a torch on each card.
4. What changed?

Then discuss

Why couldn't we see the colours in the dark?



Big idea

We need light to see colour. So does our sensor!

Teacher's program

What the computer is thinking

```
forever
    read red, green, blue amounts
    if red is the biggest → red light ON, say 'RED!'
    if green is the biggest → green light ON, say 'GREEN!'
    if blue is the biggest → blue light ON, say 'BLUE!'
```

Read it like a story

Look at the card.

Which colour came back the most?

Turn on that light!

Teacher: the program is uploaded before class (Arduino IDE 2, TCS3200/TCS34725 sensor). Children hold cards 1 to 2 cm from the sensor.

MAKE

Investigate!

Detective tests

Object	I think it will say…	It said…
Red card		
Green leaf		
Blue pen lid		
Yellow banana		
White paper		

Talk about it

What did the sensor say for yellow and white? Why do you think so?

Hold it close

Keep the object 1 to 2 cm from the sensor and very still.

Class game

Be the sorting robot

How to play

1. Put 3 boxes on the table: red, green, blue.
2. Pass each object past the sensor.
3. Put it in the box the sensor says.
4. Did the robot ever make a mistake?

Then discuss

Which objects confused the sensor? Why?



Big idea

Sensors are fast, but they can be fooled. Testing helps us find mistakes.

MAKE

Level up!

Detective challenges

Choose your challenge

- **Yellow:** what does the sensor say for yellow? Can we teach it?
- **Dark test:** cover the sensor. What happens?
- **Rainbow hunt:** find 6 colours in the classroom and test them.
- **Speed sort:** sort 10 objects in under 1 minute!

Think like an engineer

Why do sensors need light to see colour?

Boss level

Draw a sorting robot that could help in our school.

Objective check

Can I really do it? Prove it.

01 Why does a banana look yellow? ○ ○ ○

02 Show the sensor naming red, green and blue. ○ ○ ○

03 Show how your team sorted objects into boxes. ○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

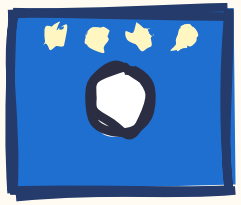
Teacher: Detective badges for every team that sorts all objects correctly!

Show what you know

What did we learn?

- [x] We see colour when light bounces off things
- [x] The sensor shines light and measures what comes back
- [x] The biggest colour wins!
- [x] Sorting robots use colour sensors

At home
Sort your toys or socks by colour. You are a sorting robot!



Exit ticket: why can't we see colours in the dark?

Next class: Crank-Powered Light!